

Life-Link

Proactive Maternal Risk Triage &
Rapid Donor Network Activation

Rural Maternal Emergency Response System — February 28, 2026

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Problem & Impact



The Crisis

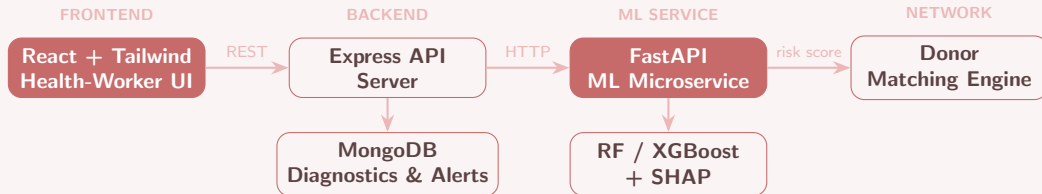
- Rural maternal emergencies **lose the golden hour** due to delayed triage
- **Postpartum hemorrhage (PPH)** is the #1 cause of maternal death globally
- Current systems are **reactive, not proactive** — donors reached too late



Our Goal

1. **Proactive risk triage** using vital-sign ML before emergencies escalate
2. **Rapid donor-network activation** matched to blood type and proximity
3. **Explainable, bias-aware** decision support for frontline workers

Impact target: Reduce PPH mortality in rural clinics by enabling *minutes-not-hours* donor response time.



Frontend

React + Tailwind — vitals entry, alerts, donor status.



Backend

Express + MongoDB for diagnostics, alerts, donor matching.



ML Microservice

FastAPI with RF/XGBoost + SHAP as standalone service.



Input & Preprocessing

- **Features:** Age, Blood Pressure, Glucose, Temperature, Heart Rate
- **Normalisation:** per-unit min-max scaling
- **Outlier clipping:** IQR-based — preserves extreme clinical signals



Bias Mitigation

- **Age-group sample weighting** at training time
- Per-group accuracy reported in model card
- SHAP global + local explanations surfaced to clinician

Clinical Safety Safeguard

Override Rule:

If *any* vital breaches extreme threshold (e.g. BP < 70 mmHg, HR > 140 bpm), system **ignores model output** and flags:

HIGH RISK !

Ensures no life-critical case is under-triaged.

Model selection: RF and XGBoost evaluated by F1-score; better performer auto-deployed per run.

Donor Profiles

- **GeoJSON** location for radius queries
- Full **medical-log timeline** per donor
- 10 000+ seeded donor records for benchmarking

Matching Filters

1. **Blood compatibility** — ABO/Rh matrix
2. **Radius proximity** — GeoJSON \$nearSphere
3. **Haemoglobin** ≥ 12.5 g/dL
4. **Disqualifying conditions** excluded

Risk score $\geq$ threshold — **Trigger alert**

Query donors by blood type + radius

Filter Hb ≥ 12.5 & exclude conditions

Return **masked** list to health worker

Admin reveals contact on confirm

 **Privacy:** Donors masked by default. Full contact revealed by authorised admins on dispatch confirmation only.

Delivered Artefacts

- **Model report (PDF):** F1-score, confusion matrix, SHAP feature importance
- **Donor seed:** 10 000+ synthetic donors with GeoJSON coordinates
- **Scalability benchmark:** query latency vs. donor pool size
- **Bias card:** per-age-group accuracy breakdown

Next Steps

1. **Hospital integrations** — HL7 FHIR API connectors for EHR ingestion
2. **SMS / WhatsApp dispatch** — Twilio / Meta Business API
3. **Audit trails** — immutable event log for every triage decision
4. **Active learning loop** — clinician feedback retrains model



Life-Link — Saving the Golden Hour, One Algorithm at a Time

